The documentation and process conversion measures necessary to comply with this document shall be completed by 28 November 2006.

INCH-POUND

MIL-PRF-19500/455F
28 August 2006
SUPERSEDING
MIL-PRF-19500/455E
25 December 2003

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER SWITCHING,
TYPES 2N5664, 2N5665, 2N5666, 2N5666S, 2N5666U3, 2N5667, AND 2N5667S,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors for use in high-speed power-switching applications. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each un-encapsulated device type as specified in MIL-PRF-19500.

* 1.2 Physical dimensions. See figure 1 (TO-66), figure 2 (TO-5 and TO-39), figure 3 (surface mount), and figure 4 (JANHC, JANKC).

* 1.3 Maximum ratings T_A = +25°C unless otherwise specified.

Type	P_T T_A = +25°C (1)	P_T T_C = +100°C (1)	$R_{\theta JA}$ (2)	$R_{\theta JC}$ (2)	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{stg} and T_J
	W	W	°C/W (max)	°C/W (max)	V dc	V dc	V dc	A dc	A dc	T_C°C
2N5664	2.5	30	70	2.6	250	200	6	5	1	
2N5665	2.5	30	70	2.6	400	300	6	5	1	
2N5666, S	1.2	15	145	6.7	250	200	6	5	1	-65 to +200
2N5666U3	1.5	35	116	2.3	250	200	6	5	1	
2N5667, S	1.2	15	145	6.7	400	300	6	5	1	

(1) For derating, see figures 5, 6, 7, 8, and 9.
(2) For thermal impedance see figures 10, 11, 12, and 13.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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1.4 Primary electrical characteristics at T_A = +25°C.

Limits	h_{FE} V_{CE} = 5 V I_C = 1 A		$\|h_{fe}\|$ V_{CE} = 5 V I_C = 0.5 A dc f = 10 MHz	$V_{BE(sat)}$ I_C = 3 A dc (1)	$V_{CE(sat)}$ I_C = 3 A dc (1)	Pulse response		
						t_{on} I_C = 1 A dc	t_{off} I_C = 1 A dc	
	2N5665 2N5667, S	2N5664 2N5666, S, U3					2N5664 2N5666, S, U3	2N5665 2N5667, S
				V dc	V dc	µ s	µ s	µ s
Min	25	40	2.0					
Max	75	120	7.0	1.2	0.4	0.25	1.5	2.0

(1) I_B = 0.3 A dc for 2N5664, 2N5666, 2N5666S, 2N5666U3; I_B = 0.6 A dc for 2N5665, 2N5667, 2N5667S.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

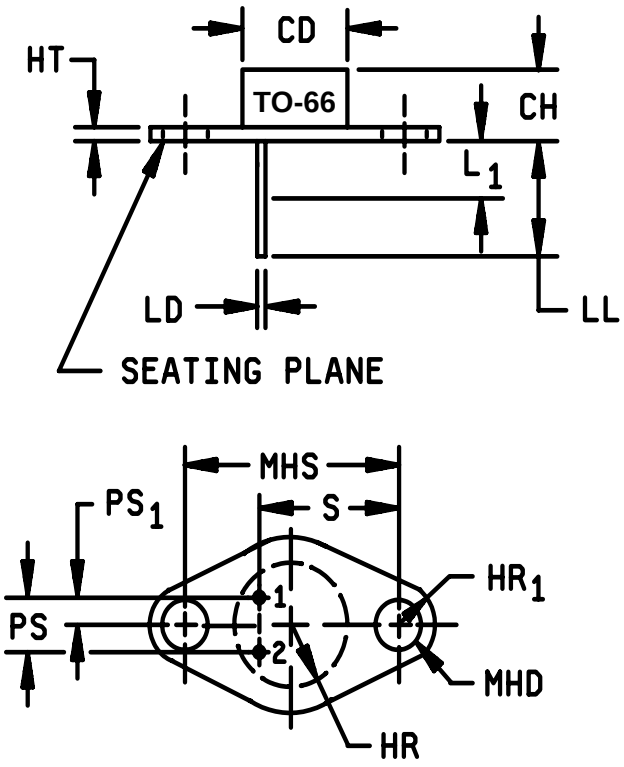
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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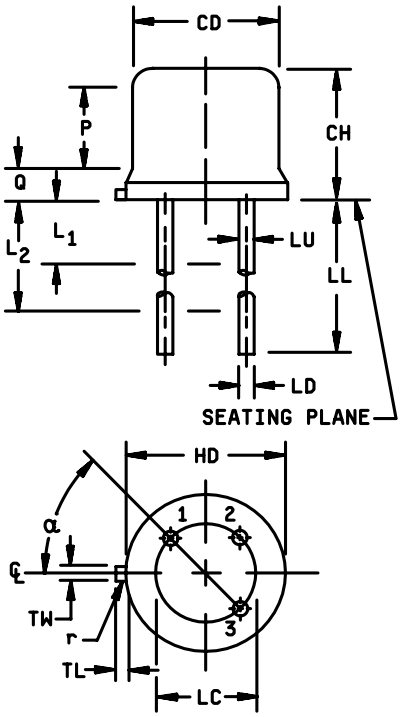


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.620		15.75	3
CD_1	.470	.500	11.94	12.70	
CH	.250	.340	6.35	8.64	3
HR		.350		8.89	6
HR_1	.115	.145	2.92	3.68	
HT	.050	.075	1.27	1.91	3
HT_1		.050		1.27	3
LD	.028	.034	.711	.863	5, 9
LL	.360	.500	9.14	12.70	5, 9
L_1		.050		1.27	4
MHD	.142	.152	3.62	3.86	7
MHS	.958	.962	24.33	24.43	
PS	.190	.210	4.83	5.33	4
PS_1	.093	.107	2.36	2.72	4
S_1	.570	.590	14.48	14.99	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Body contour is optional within zone defined by LD and CD.
4. These dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. LD applies between L_1 and LL. Diameter is uncontrolled in L_1.
10. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Physical dimensions of transistor types 2N5664 and 2N5665.

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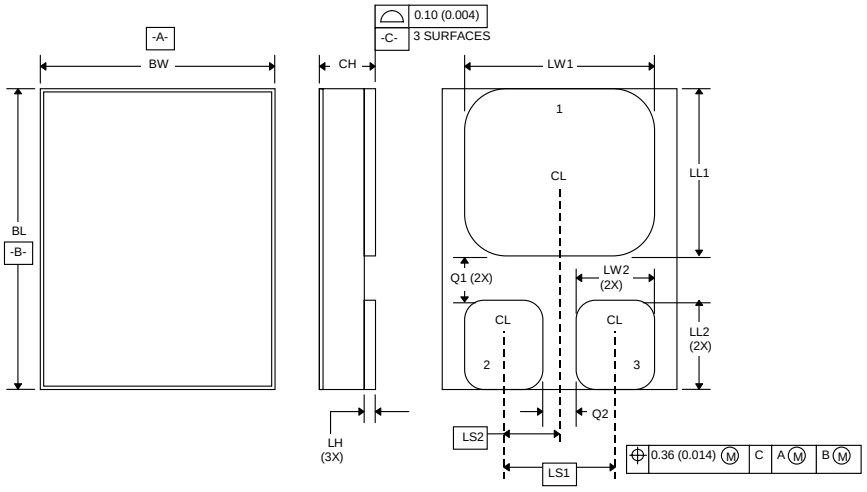
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	3
LL	See notes 13 and 14				
L_1		.050		1.27	10
L_2	.250		6.35		10
LU	.016	.019	0.41	0.48	4
P	.100		2.54		5
Q					6
r		.007		0.18	
α	45° TP		45° TP		7
TL	.029	.045	0.74	1.14	
TW	.028	.034	0.71	0.86	

NOTES:

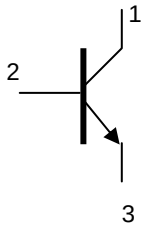
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Measured in the zone beyond .250 inches (6.35 mm) from the seating plane.
4. Measured in the zone .050 inches (1.27 mm) and .250 inches (6.35 mm) from the seating plane.
5. Variations on dimension CD in this zone shall not exceed .010 inches (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane .054 inches +.001, -.000 (1.37 mm +.03, -.00) below the seating plane of the transistor, maximum diameter leads shall be within .007 inches (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads
11. Diameter of leads in this zone is not controlled.
12. Lead 1 - Emitter; lead 2 - Base, lead 3 - Collector.
13. For transistor types 2N5666 and 2N5667, LL is 1.500 inches (38.1 mm) minimum and 1.75 inches (44.45 mm) maximum.
14. For transistor types 2N5666S and 2N5667S, LL is .500 inches (12.7 mm) minimum and .75 inches (19.05 mm) maximum.
15. * In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 2. Physical dimensions of transistor types 2N5666, 2N5666S, 2N5667 and 2N5667S (TO-5 and TO-39).

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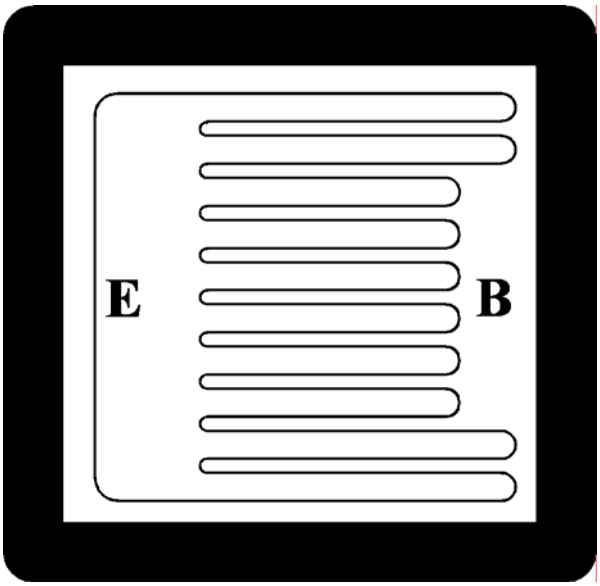


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.395	.405	10.04	10.28
BW	.291	.301	7.40	7.64
CH	.1085	.1205	2.76	3.06
LH	.010	.020	0.25	0.51
LW1	.281	.291	7.14	7.41
LW2	.090	.100	2.29	2.54
LL1	.220	.230	5.59	5.84
LL2	.115	.125	2.93	3.17
LS1	.150 BSC		3.81 BSC	
LS2	.075 BSC		1.91 BSC	
Q1	.030		0.762	
Q2	.030		0.762	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
4. Terminal 1 - collector, terminal 2 - base, terminal 3 - emitter.

FIGURE 3. Physical dimensions, surface mount (2N5666U3 version).

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1. Chip size: .120 x .120 inch ± .002 inch (3.05 mm x 3.05 mm ±0.051 mm).
2. Chip thickness: .010 ± .0015 inch nominal, (2.54 mm ±0.038 mm).
3. Top metal: Aluminum 30,000Å minimum, 33,000Å nominal
4. Back metal: A. Al/Ti/Ni/Ag12kÅ/3kÅ/7kÅ/7kÅmin.15kÅ/5kÅ/10kÅ/10kÅ nominal.
   B. Gold 2,500 Å minimum, 3,000Å nominal.
5. Backside: Collector
6. Bonding pad: B = .052 x .012 inch (1.32 mm x 0.305 mm),
   E = .084 x .012 mils (2.13 mm x 0.305 mm).

FIGURE 4. JANHC and JANKC (A-version) die dimensions.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-66), figure 2 (TO-5 and TO-39), figure 3 (surface mount), and figure 4 (JANHC, JANKC).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

* 3.6 Electrical test requirements. The electrical test requirements shall be as specified in 4.4.2, 4.4.3, and 4.4.4, herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (list applicable JAN levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance method 3131 of MIL-STD-750, see 4.3.3.	Thermal impedance method 3131 of MIL-STD-750, see 4.3.3.
7	Optional	Optional
9	I_{CES1} and h_{FE2}	I_{CES1}
11	I_{CES1} and $h_{FE2;}$ ΔI_{CES1} = 100 percent of initial value or 10 nA dc, whichever is greater; Δh_{FE2} = ± 15 percent.	I_{CES1} and $h_{FE2;}$ ΔI_{CES1} = 100 percent of initial value or 20 nA dc, whichever is greater.
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CES1} = +100 percent of initial value or 10 nA dc, whichever is greater. Δh_{FE2} = ±15 percent.	Subgroup 2 of table I herein; ΔI_{CES1} = +100 percent of initial value or 20 nA dc, whichever is greater. Δh_{FE2} = ±25 percent.
14	Required	Required

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_J = +187.5 ±12.5°C, V_{CE} = 100 V dc, $T_A \leq$ +55°C. Burn-in duration for lot acceptance for the JANKC level follows JANS requirements. Burn-in duration for lot acceptance for the JANHC level follows JANTX requirements.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed to insure the assembled chips will meet the requirements of table I, subgroup 2.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed on each die in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, (and V_C where appropriate). Measurement delay time (t_{MD}) = 70 μS maximum. The thermal impedance limit used in screen 3c of 4.3 and the subgroup 2 of table I herein shall comply with the thermal impedance graph on figures 10, 11, 12, and 13 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131. See Group E subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with table I, subgroup 2.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta measurements shall be in accordance with the steps in table II herein as specified in the notes for table II.

4.4.2.1 Group B inspection, table E-VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 30 V dc minimum, T_A = +25°C ± 3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the heating cycle shall be permitted.
B5	1027	See 4.5.3 herein.
B6	3131	See 1.3 and 4.3.3 herein.

4.4.2.2 Group B inspection, table E-VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	T_J = +187.5°C ± 12.5°C, V_{CE} = 100 ± 5 V dc; T_A = ≤ +100°C.
B5	3131	See 1.3 and 4.3.3 herein.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta measurements shall be in accordance with the steps in table II herein as specified in the notes for table II.

Subgroup	Method	Condition
C2	2036	Terminal strength (tension) 2N5664 and 2N5665 only: Test condition A, weight = 3 pounds, application time = 15 seconds. Terminal strength (lead fatigue) 2N5666, 2N5666S, 2N5667 and 2N5667S only: Test condition E (Not applicable to 2N5666U3).
C6	1027	2N5664, 2N5665, 2N5666S, 2N5666U3,: T_C = +100°C; P_T = 30 W; V_{CE} = 30 V dc. 2N5666, 2N5666S, 2N5666U3, 2N5667 and 2N5667S, T_A = +25°C; P_T = 1.2 W; V_{CE} = 40 V.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; Delta measurements shall be in accordance with the applicable steps of table II.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted at a case temperature (T_C) of +25°C.

4.5.3 Group B accelerated life test. This test shall be conducted using one of the four options listed herein (a, b, c, or d) with the following conditions applying to all options: V_{CB} = 30 V dc.

a. T_A = +150°C, maximum, t = 96 hours minimum, T_J = +275°C.

b. P_T = 2.5 W (TO-66); P_T = 1.2 W (TO-5, U3 suffix), T_A = +112°C or P_T adjusted to give a lot average of T_J = +275°C, t = 96 hours minimum, T_J = +275°C.

c. T_A = +25°C +3 °C with P_T adjusted to give a lot average of T_J = +275°C, t = 96 hours minimum, T_J = +275°C.

d. Adjust T_a or P_d to achieve a T_J = +225°C minimum. t = 216 hours, n = 45, c = 0.

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* TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
Breakdown voltage collector to emitter	3011	Bias condition B; I_C = 10 mA dc, pulsed (see 4.5.1), R_1 = 100 Ω	$V_{(BR)CER}$			
2N5664, 2N5666, 2N5666S				250		V dc
2N5665, 2N5667, 2N5667S				400		V dc
Breakdown voltage emitter to base	3026	Bias condition D, I_E = 10 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	6		V dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES1}		0.2	µA dc
2N5664, 2N5666, 2N5666S		V_{CE} = 200 V dc				
2N5665, 2N5667, 2N5667S		V_{CE} = 300 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N5664, 2N5666, 2N5666S		V_{CB} = 200 V dc			0.1	µA dc
2N5664, 2N5666, 2N5666S		V_{CB} = 250 V dc			1.0	mA dc
2N5665, 2N5667, 2N5667S		V_{CB} = 300 V dc			0.1	µA dc
2N5665, 2N5667, 2N5667S		V_{CE} = 400 V dc			1.0	mA dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 0.5 A dc pulsed (see 4.5.1)	h_{FE1}			
2N5664, 2N5666, 2N5666S				40		
2N5665, 2N5667, 2N5667S				25		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1.0 A dc pulsed (see 4.5.1)	h_{FE2}			
2N5664, 2N5666, 2N5666S				40	120	
2N5665, 2N5667, 2N5667S				25	75	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 3.0 A dc pulsed (see 4.5.1)	h_{FE3}			
2N5664, 2N5666, 2N5666S				15		
2N5665, 2N5667, 2N5667S				10		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 5 A dc, pulsed (see 4.5.1)	h_{FE4}	5		
Collector-emitter saturation voltage	3071	I_C = 3.0 A dc,	$V_{CE(sat)1}$		0.4	V dc
2N5664, 2N5666, 2N5666S		I_B = 0.3 A dc, pulsed (see 4.5.1)				
2N5665, 2N5667, 2N5667S		I_B = 0.6 A dc, pulsed (see 4.5.1)				
Collector-emitter saturation voltage	3071	I_C = 5 A dc, I_B = 1 A dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$ 3/		1.0	V dc
Base-emitter saturation voltage	3066	Test condition A, I_C = 3.0 A dc,	$V_{BE(sat)1}$ 3/		1.2	V dc
2N5664, 2N5666, 2N5666S		I_B = 0.3 A dc, pulsed (see 4.5.1)				
2N5665, 2N5667, 2N5667S		I_B = 0.6 A dc, pulsed (see 4.5.1)				
Base-emitter saturation voltage	3066	Test condition A, I_C = 5 A dc, I_B = 1 A dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.5	V dc
Subgroup 3						
High-temperature operation:		T_A = + 150°C				
Collector to emitter cutoff current	3041	Bias condition C	I_{CES2}		100	µA dc
2N5664, 2N5666, 2N5666S		V_{CE} = 200 V dc				
2N5665, 2N5667, 2N5667S		V_{CE} = 300 V dc				
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1.0 A dc, pulsed (see 4.5.1)	h_{FE5}			
2N5664, 2N5666, 2N5666S				15		
2N5665, 2N5667, 2N5667S				10		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Magnitude of common-emitter, small-signal short-circuit, forward-current, transfer ratio	3306	V_{CE} = 5 V dc, I_C = 0.5 A dc f = 10 MHz	$\lvert h_{fe} \rvert$	2.0	7.0	
Open-circuit output capacitance	3236	V_{CB} = 10 V dc, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		120	pF
Pulse response						
Turn-on time	3251	Test condition A; I_C = 1.0 A dc, V_{CC} = 100 V dc	t_{on}		0.25	µs
2N5664, 2N5666, 2N5666S		See figure 14				
2N5665, 2N5667, 2N5667S		See figure 15				
Turn-off time	3251	Test condition A; I_C = 1.0 A dc, V_{CC} = 100 V dc	t_{off}			
2N5664, 2N5666, 2N5666S		See figure 14			1.5	µs
2N5665, 2N5667, 2N5667S		See figure 15			2.0	µs
Subgroup 5						
Safe operating area (continuous dc) (for types 2N5664 and 2N5665 only)	3051	T_C = + 100°C, t ≥ 1 s, 1 cycle; $t_r + t_f$ = 10 µs (see figure 16)				
Test #1 2N5664 and 2N5665		V_{CE} = 6 V dc, I_C = 5 A dc				
Test #2 2N5664 and 2N5665		V_{CE} = 40 V dc, I_C = 0.75 A dc				
Test #3 2N5664		V_{CE} = 200 V dc, I_C = 43 mA dc				
Test #4 2N5665		V_{CE} = 300 V dc, I_C = 21 mA dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (continuous dc) (for types 2N5666, 2N5666S, 2N5667, and 2N5667S	3051	T_C = +100°C, t ≥ 1 s, 1 cycle; $t_r + t_f$ = 10 μs (see figure 17)				
Test #1 2N5666, 2N5666S, 2N5667, and 2N5667S		V_{CE} = 3.0 V dc, I_C = 5 A dc				
Test #2 2N5666, 2N5666S, 2N5667, and 2N5667S		V_{CE} = 37.5 V dc, I_C = 0.4 A dc				
Test #3 2N5666 and 2N5666S		V_{CE} = 200 V dc, I_C = 27 mA dc				
Test #4 2N5667 and 2N5667S		V_{CE} = 300 V dc, I_C = 14 mA dc				
Safe operating area (switching)	3053	Load condition B (clamped inductive load) (see figure 18); T_C = + 100°C, $t_r + t_f \leq 10$ μs, duty cycle ≤ 2 percent; t_p = 4 ms; R_S = 0.5 Ω, R_{BB1} = 50 Ω, V_{BB1} = 50 V dc R_{BB2} = 50 Ω, V_{BB2} = -4 V dc I_C = 5 A dc, V_{CC} = 50 V dc $R_L \leq 2.5$ Ω, L = 40 mH (Triad C-48U or equivalent)				
2N5664 2N5666 and 2N5666S		Clamp voltage = 200 +0, -5 V dc				
2N5665 2N5667 and 2N5667S		Clamp voltage = 300 +0, -5 V dc				
End-point electrical measurements		See table I, group A, subgroup 2 herein				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics for 2N5666U3 are identical to 2N5666 unless otherwise noted.
3/ Measured at less than .125 inch (3.175 mm) from case.

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TABLE II. Groups B, C, and E delta measurements. 1/ 2/ 3/

Steps	Inspection 4/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Base condition C	ΔI_{CES1}	100 percent of initial value or 20 nA dc, whichever is greater.		
	2N5664 2N5666, 2N5666S, 2N5666U3		V_{CE} = 200 V dc				
	2N5665 2N5667, 2N5667S		V_{CE} = 300 V dc				
2.	Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1.0 A dc pulsed (see 4.5.1)	Δh_{FE2} 5/	± 25 percent change from initial reading.		

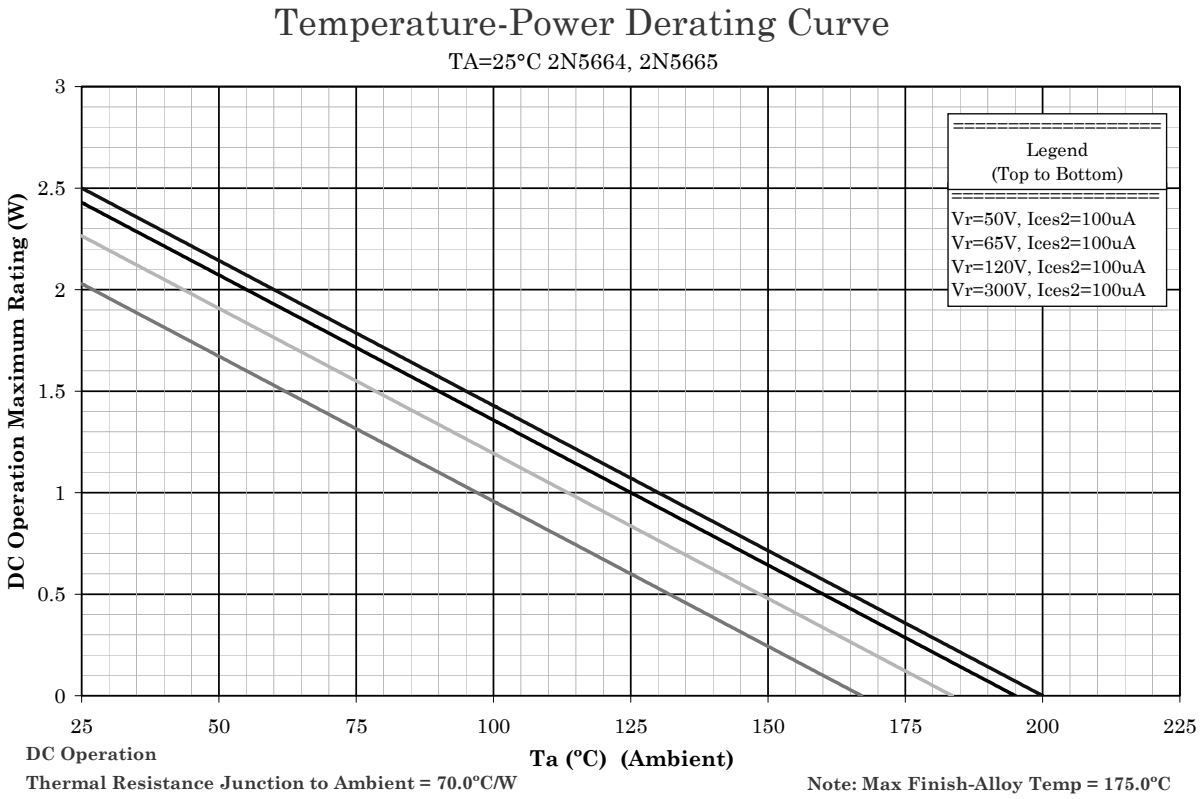
1/ The delta measurements for table E-VIa (JANS) of MIL-PRF-19500 are after subgroups 4 and 5, and consist of steps 1 and 2 of table II herein.
2/ The delta measurements for table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows: Subgroups 3 and 6, see table II herein, steps 1 and 2.
3/ The delta measurements for table E-VII of MIL-PRF-19500 are as follows: Subgroup 6, see table II herein, steps 1 and 2.
4/ See MIL-PRF-19500 for sampling plan.
5/ Measured at less than .125 inch (3.175 mm) from case.

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* TABLE III. Group E inspection (all quality levels) - for qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and table II, step 1 and 2.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles.	
Electrical measurements		See table I, subgroup 2 and table II herein.	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500.	Sample size N/A
Subgroup 5			15 devices, c = 0
Barometric pressure (reduced)	1001	Normal mounting pressure = 8 mm Hg ±2 mm Hg for 60 s (minimum)	
Electrical measurements		See table I, subgroup 2	
Subgroup 6			3 devices
ESD (electrostatic discharge)			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V, Condition B for devices < 400 V.	

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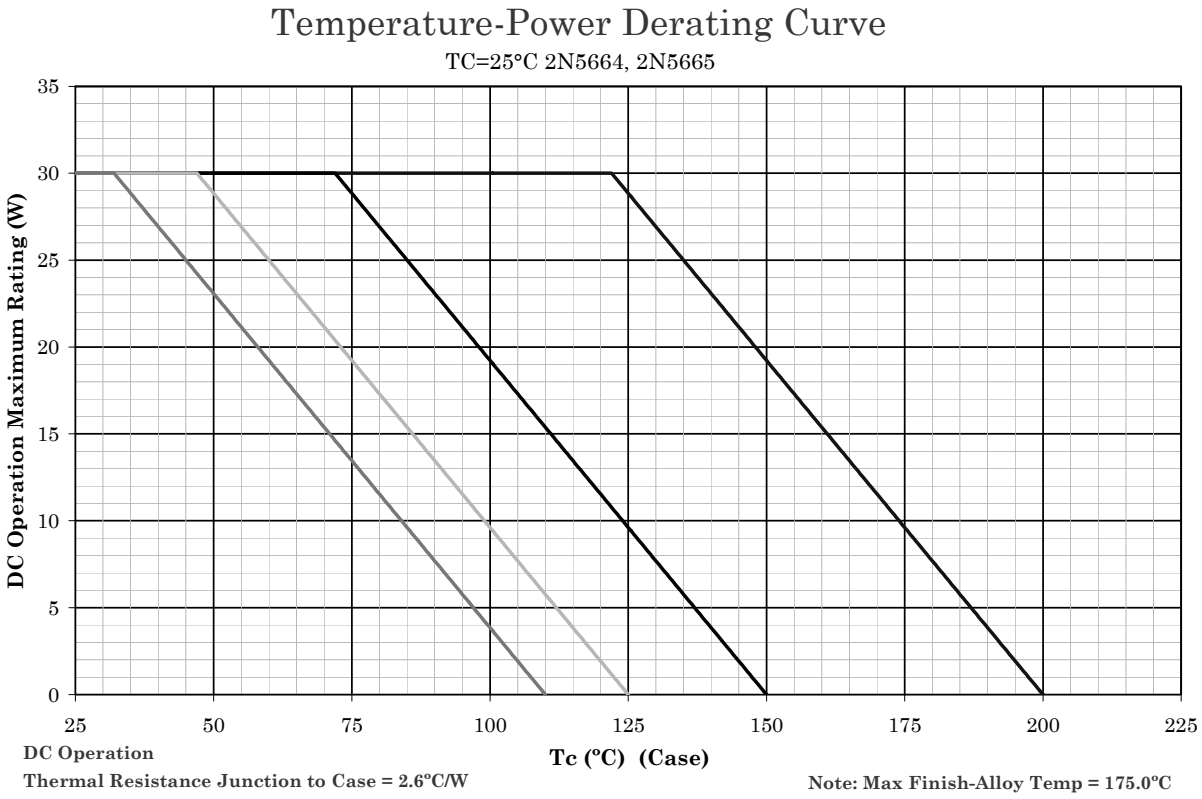


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Top derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.) and a maximum DC voltage of 50V.
3. Second derate design curve chosen for a maximum DC voltage of 65V.
4. Third derate design curve chosen for a maximum DC voltage of 120V.
5. Fourth and bottom derate design curve chosen for the maximum rated DC voltage of 300V.
6. All four curves show the predicted effects of high voltage DC operation for devices with leakage running at the Ices2 maximum rated limited with free air cooling to prevent thermal runaway. Improved cooling or lower duty cycles will, of course, increase the maximum voltage you can use as illustrated in the other case mounted derating curve sets.

* FIGURE 5. Temperature-power derating for 2N5664 and 2N5665.

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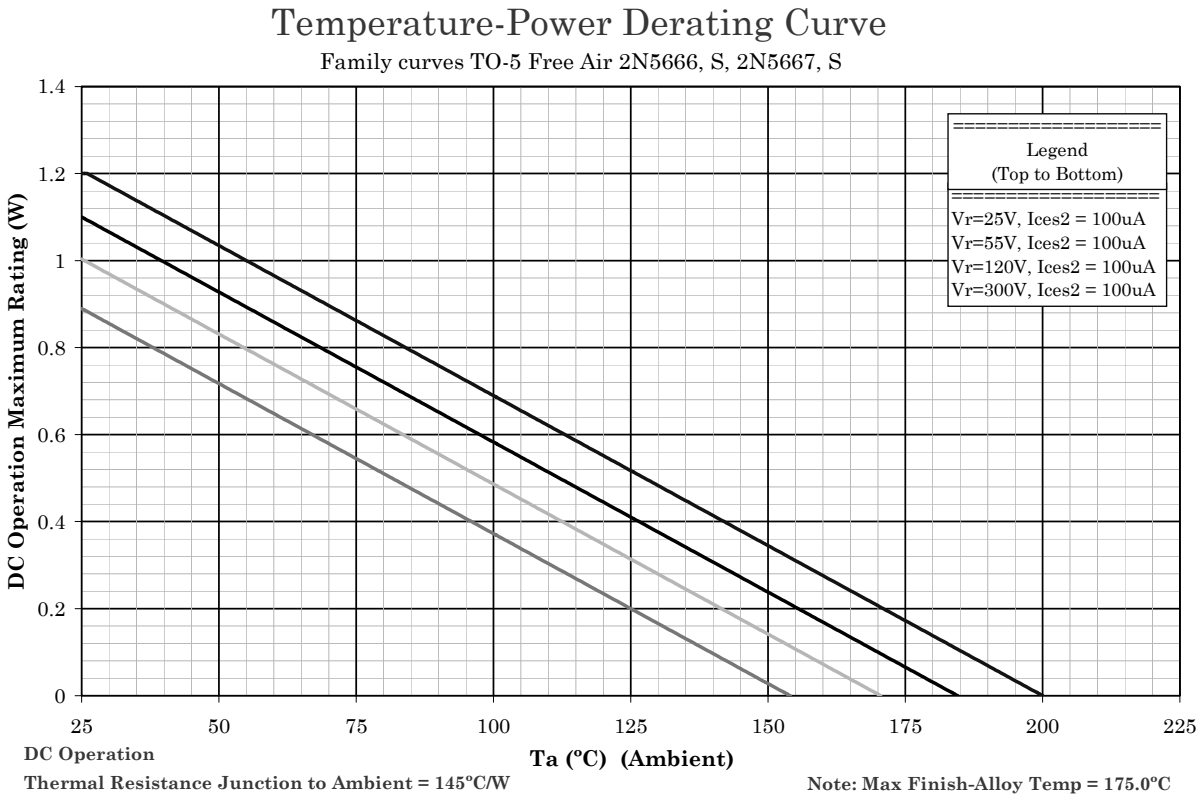


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 6. Temperature-power derating for 2N5664 and 2N5665.

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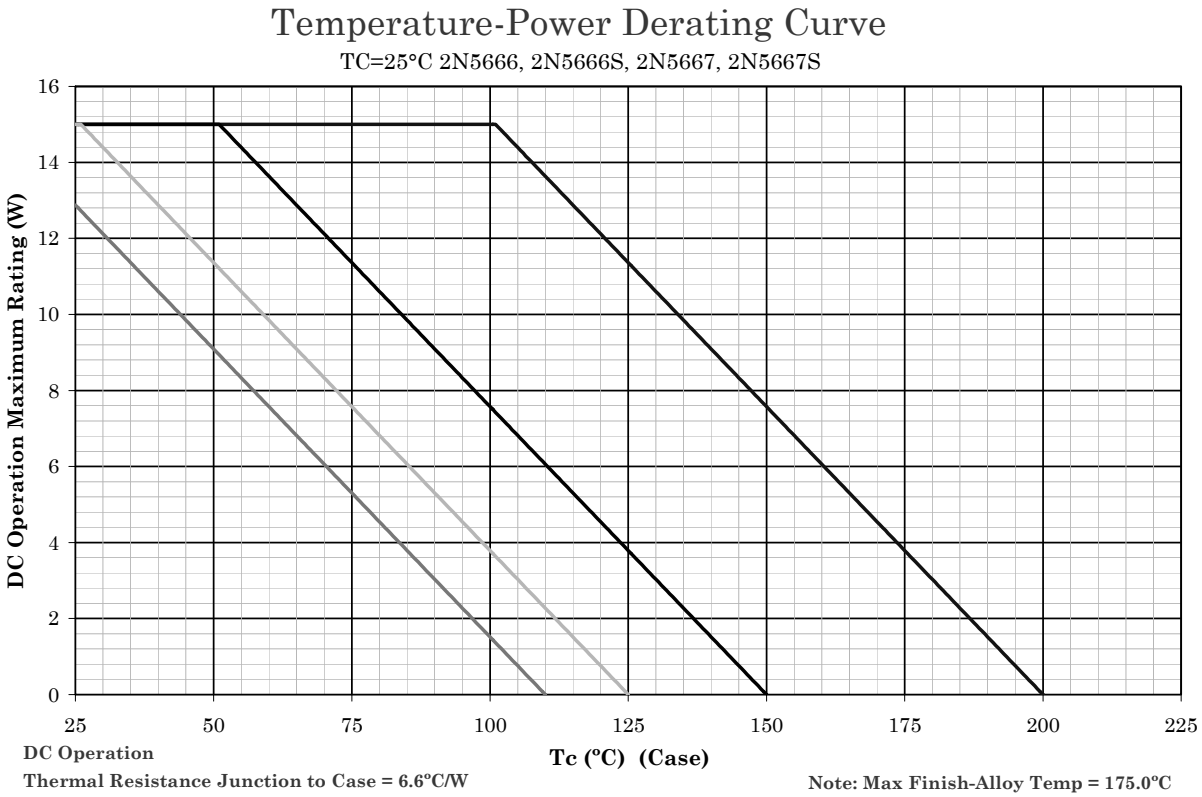


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Top derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.) and a maximum DC voltage of 25V.
3. Second derate design curve chosen for a maximum DC voltage of 55V.
4. Third derate design curve chosen for a maximum DC voltage of 120V.
5. Fourth and bottom derate design curve chosen for the maximum rated DC voltage of 300V.
6. All four curves show the predicted effects of high voltage DC operation for devices with leakage running at the Ices2 maximum rated limited with free air cooling to prevent thermal runaway. Improved cooling or lower duty cycles will, of course, increase the maximum voltage you can use as illustrated in the other case mounted derating curve sets.

* FIGURE 7. Temperature-power derating for 2N5666, 2N5666S, 2N5667, and 2N5667S.

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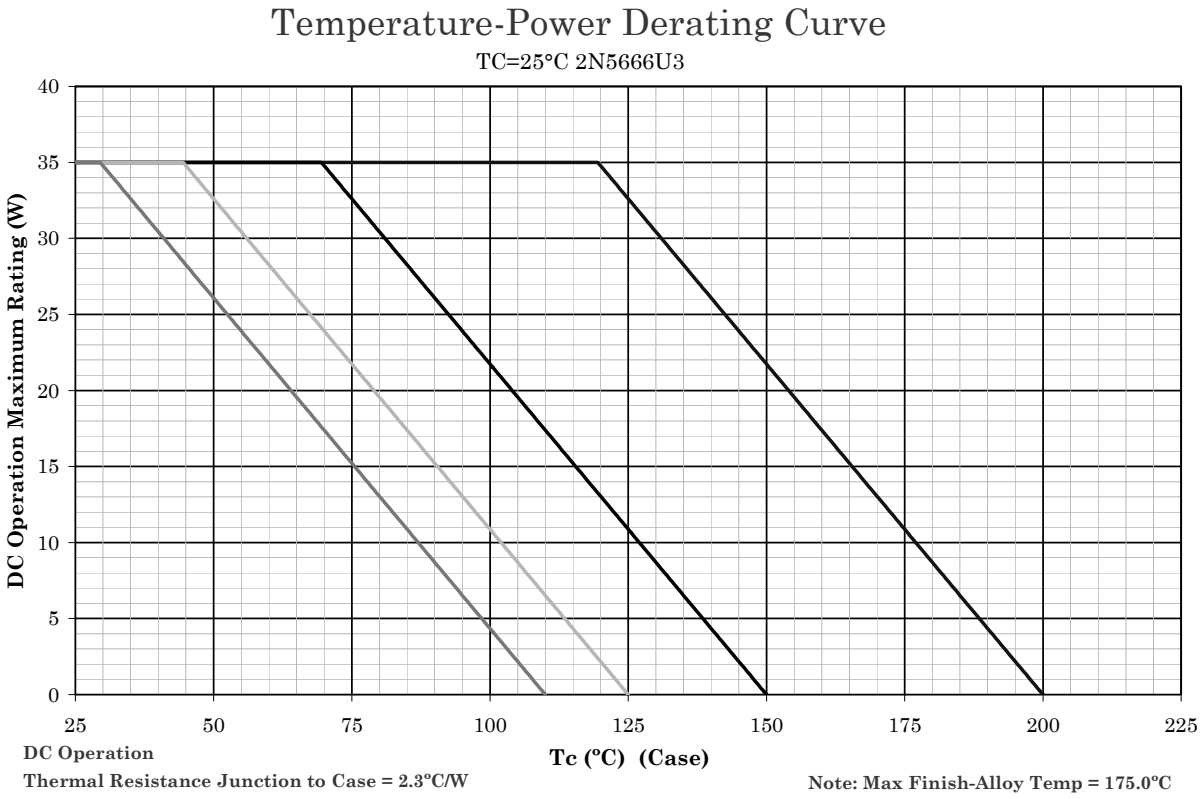


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 8. Temperature-power derating for 2N5666, 2N5666S, 2N5667, and 2N5667S.

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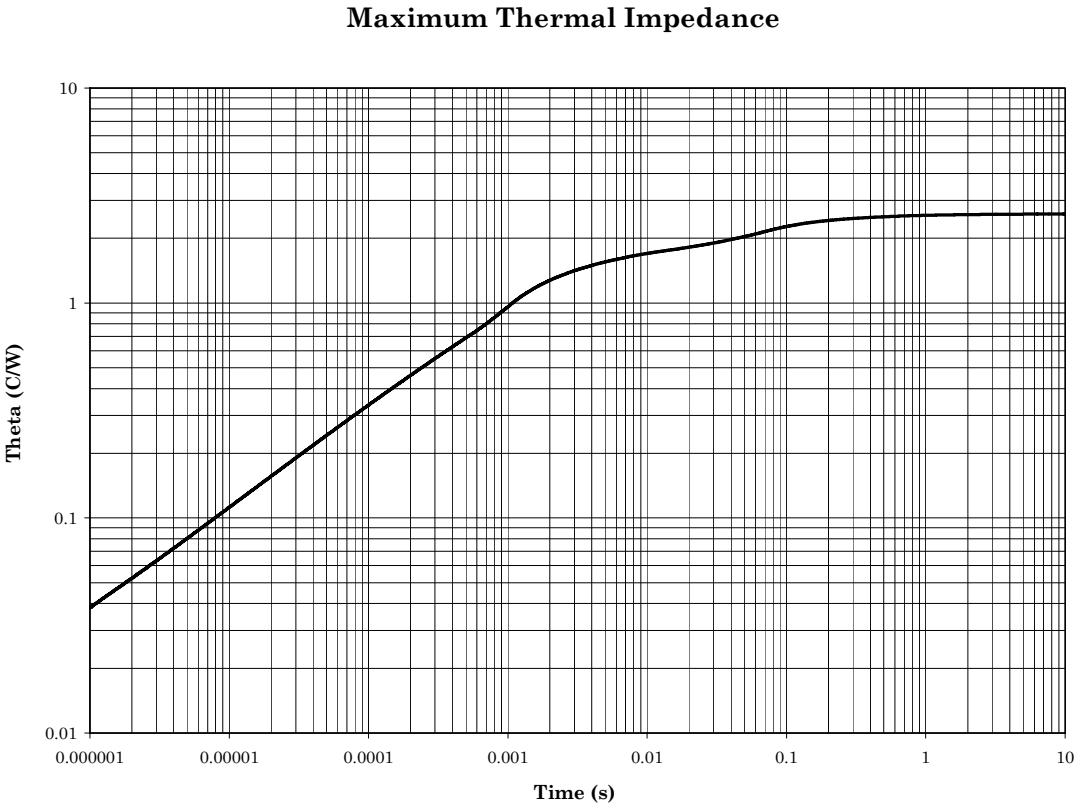


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 9. Temperature-power derating for 2N5666U3.

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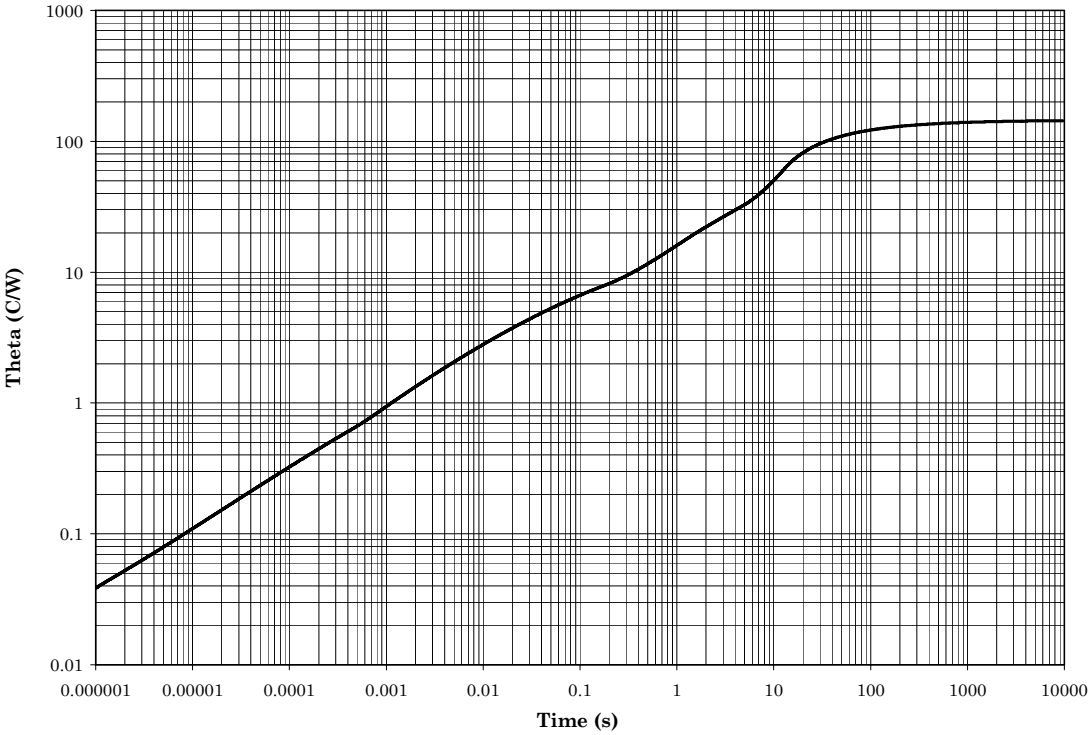


Solder mounted to copper heatsink at T_C = +25°C, thermal resistance = 2.6°C/W.

FIGURE 10. Thermal impedance graph ($R_{\theta JC}$) for 2N5664, and 2N5665 (TO-66).

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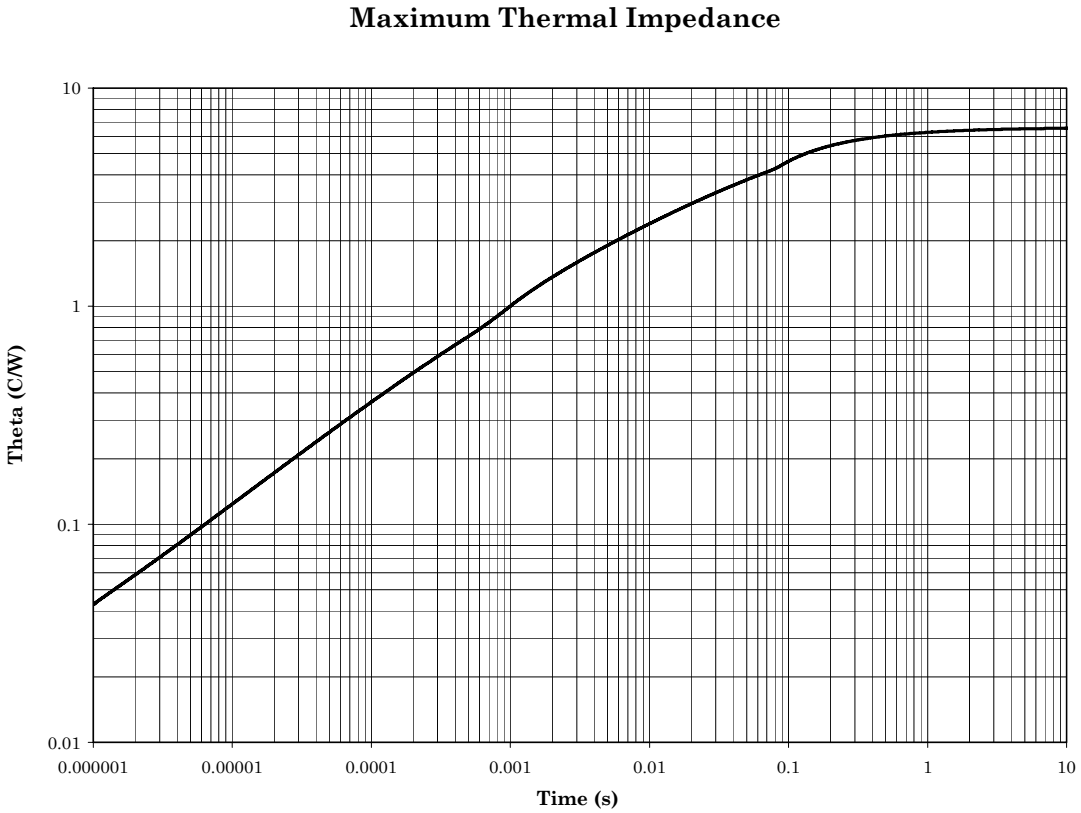
Maximum Thermal Impedance



T_A = +25°C, thermal resistance $R_{\theta JA}$ = 145°C/W (ambient thermal resistance varies with power).

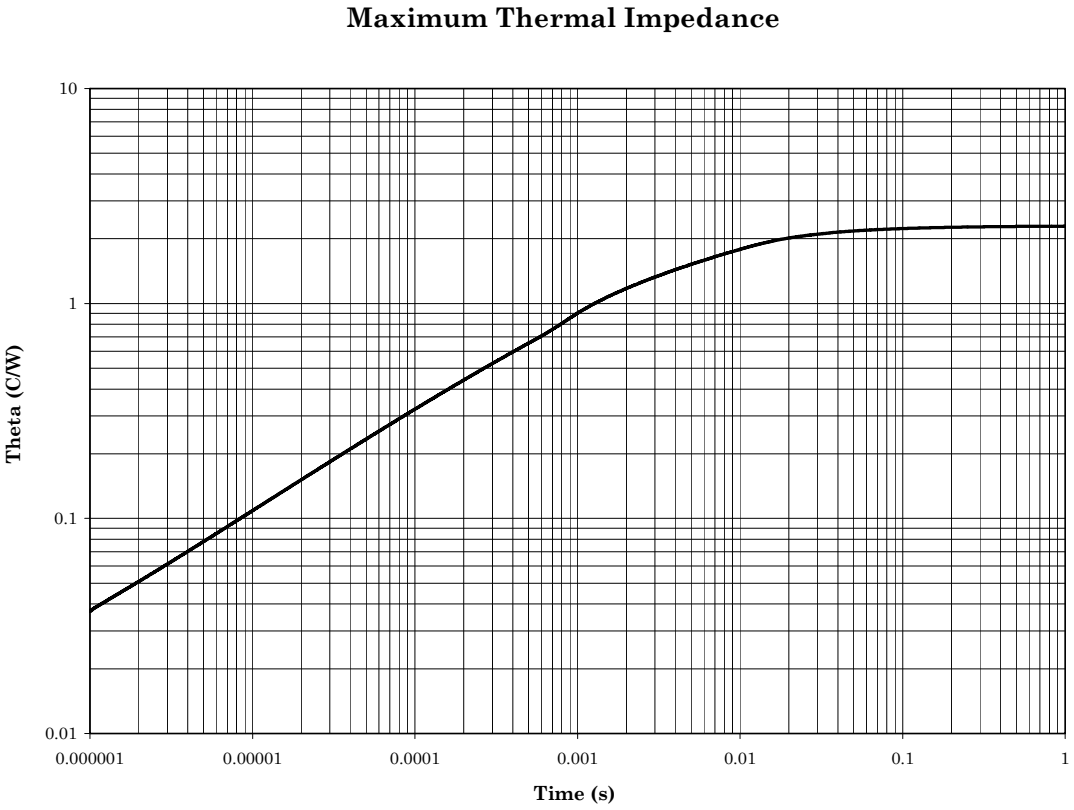
* FIGURE 11. Thermal impedance graph ($R_{\theta JA}$) for 2N5666, 2N5666S, 2N5667, and 2N5667S (TO-5 and TO-39).

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* FIGURE 12. Thermal impedance graph ($R_{\theta JC}$) for 2N5666, 2N5666S, 2N5667 and 2N5667S.

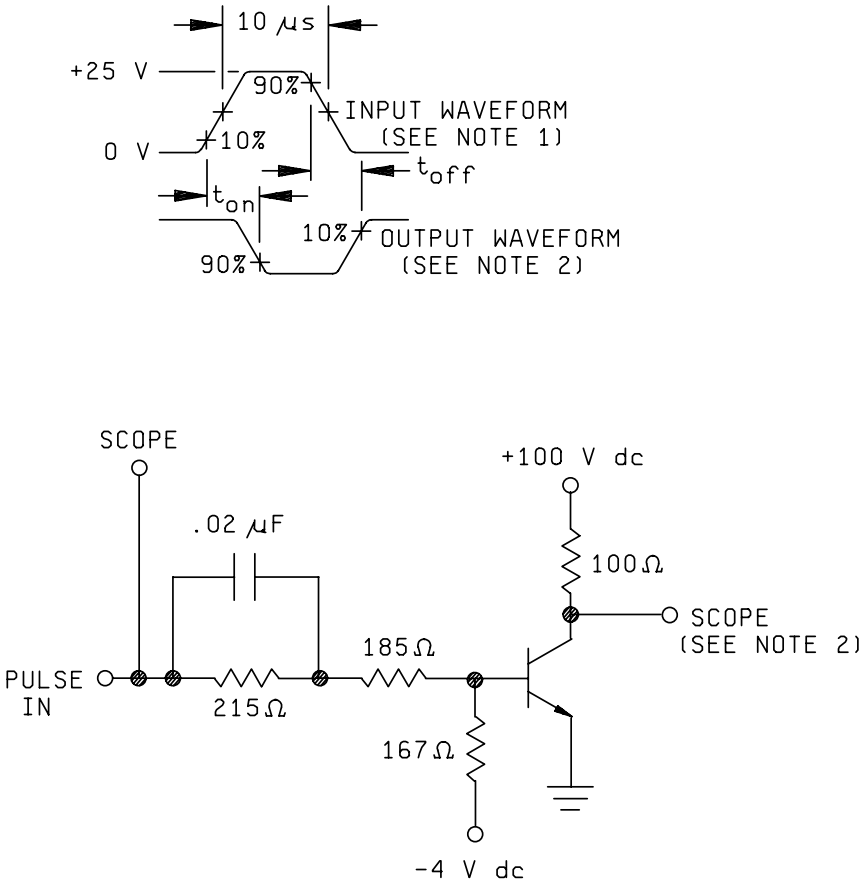
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Solder mounted to copper heatsink at T_C = +25°C, thermal resistance $R_{\theta JC}$ = 2.3°C/W.

* FIGURE 13. Thermal impedance graph ($R_{\theta JC}$) for 2N5666U3.

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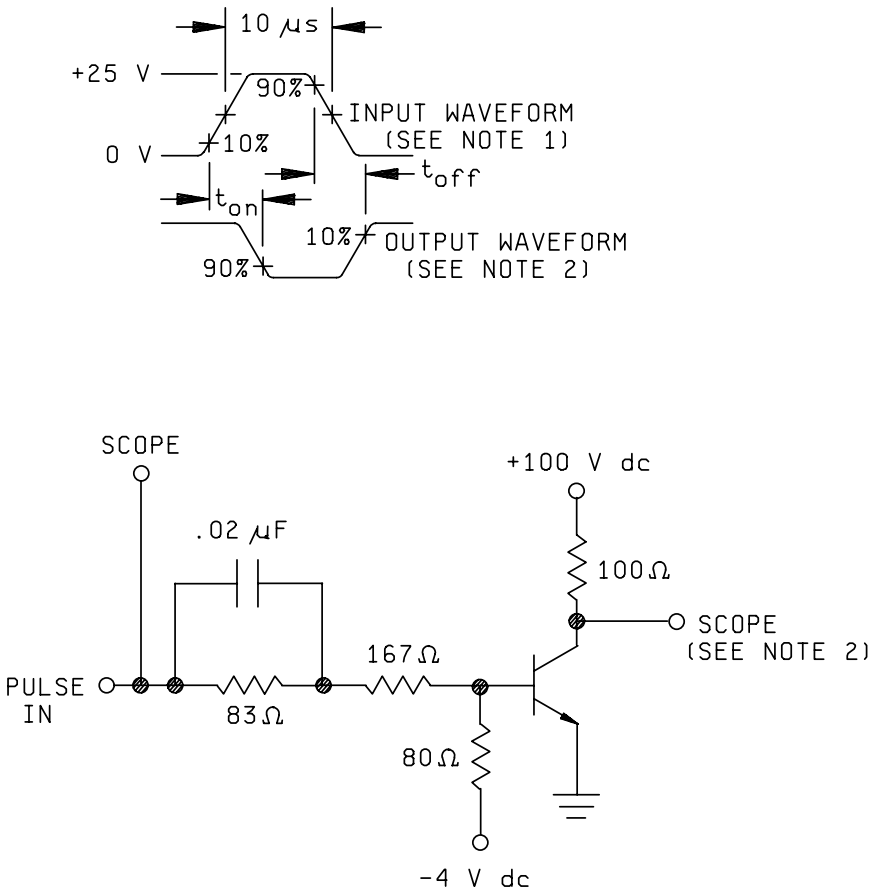


NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{out} = 50$ ohm, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $Z_{in} \geq 10$ M Ω, $C_{in} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages and supply voltages (-4V dc and +100 V dc) are nominal and shall be adjusted to obtain $I_{B1} = - I_{B2} = 30$ mA and $I_C = 1$ A.
6. An equivalent circuit may be used.
7. 0.02 µF capacitor may be removed during voltage adjustments.

FIGURE 14. Pulse response test circuit for types 2N5664, 2N5666, 2N5666S, and 2N5666U3.

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NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, Z_{out} = 50 ohm, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $Z_{in} \geq 10$ MΩ, $C_{in} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages and supply voltages (-4 V dc and +100 V dc) are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 50$ mA and $I_C = 1$ A.
6. An equivalent circuit may be used.
7. 0.02 µF capacitor may be removed during voltage adjustments.

FIGURE 15. Pulse response test circuit for types 2N5665, 2N5667 and 2N5667S.

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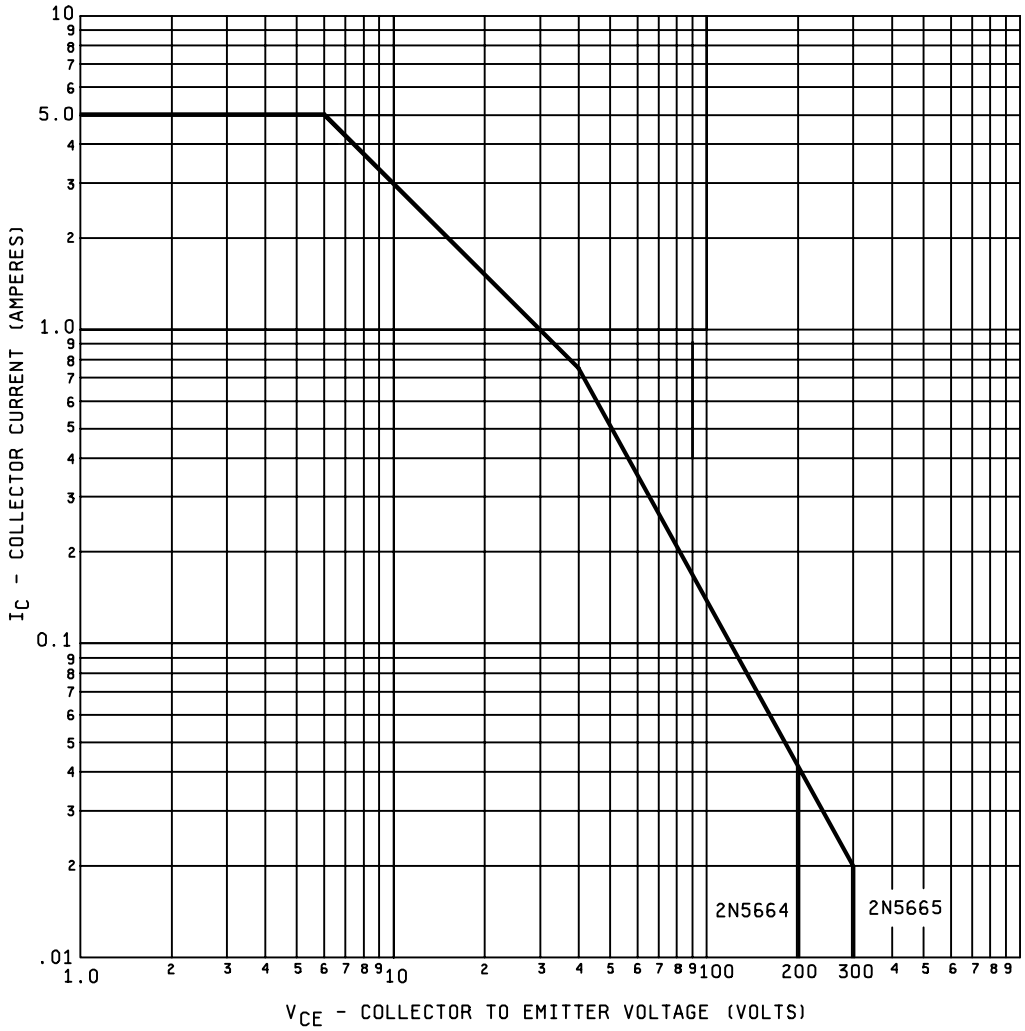
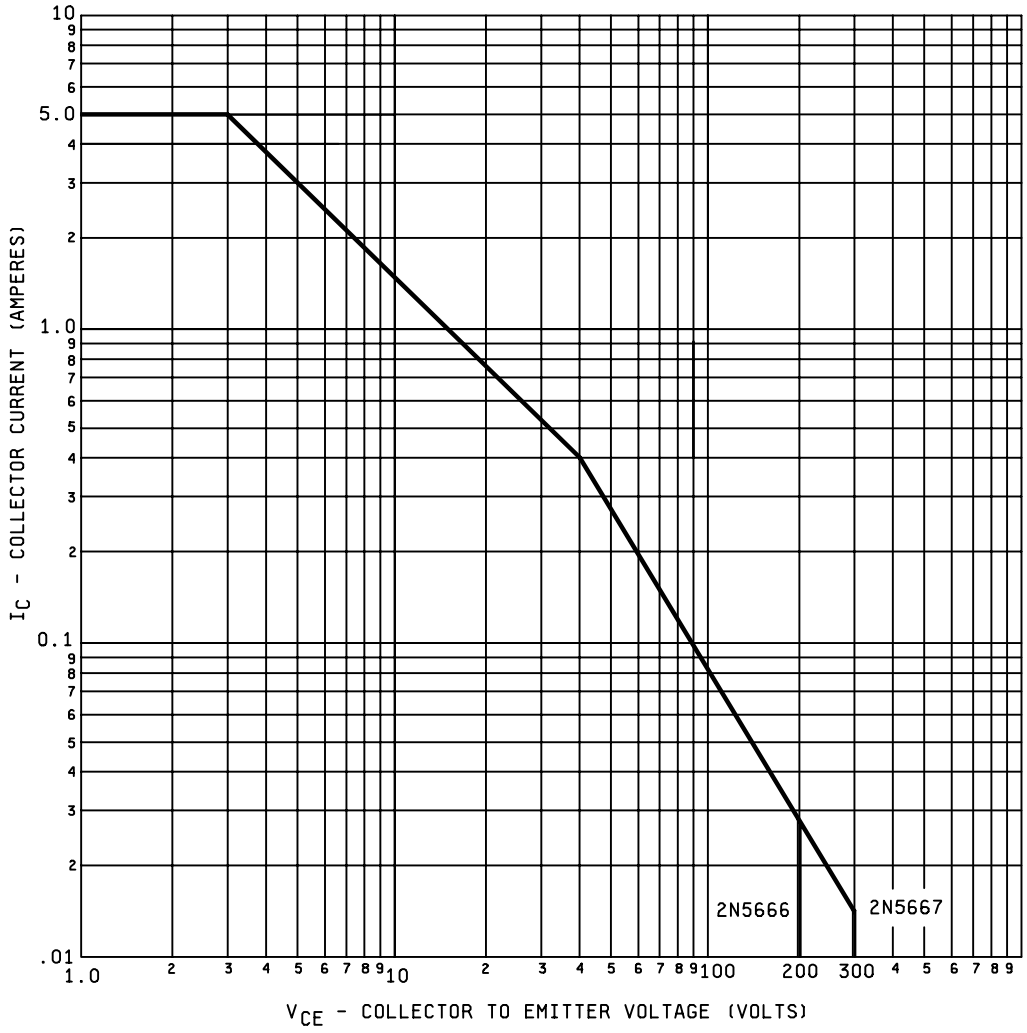


FIGURE 16. Maximum safe operating graph (continuous dc) for types 2N5664 and 2N5665.

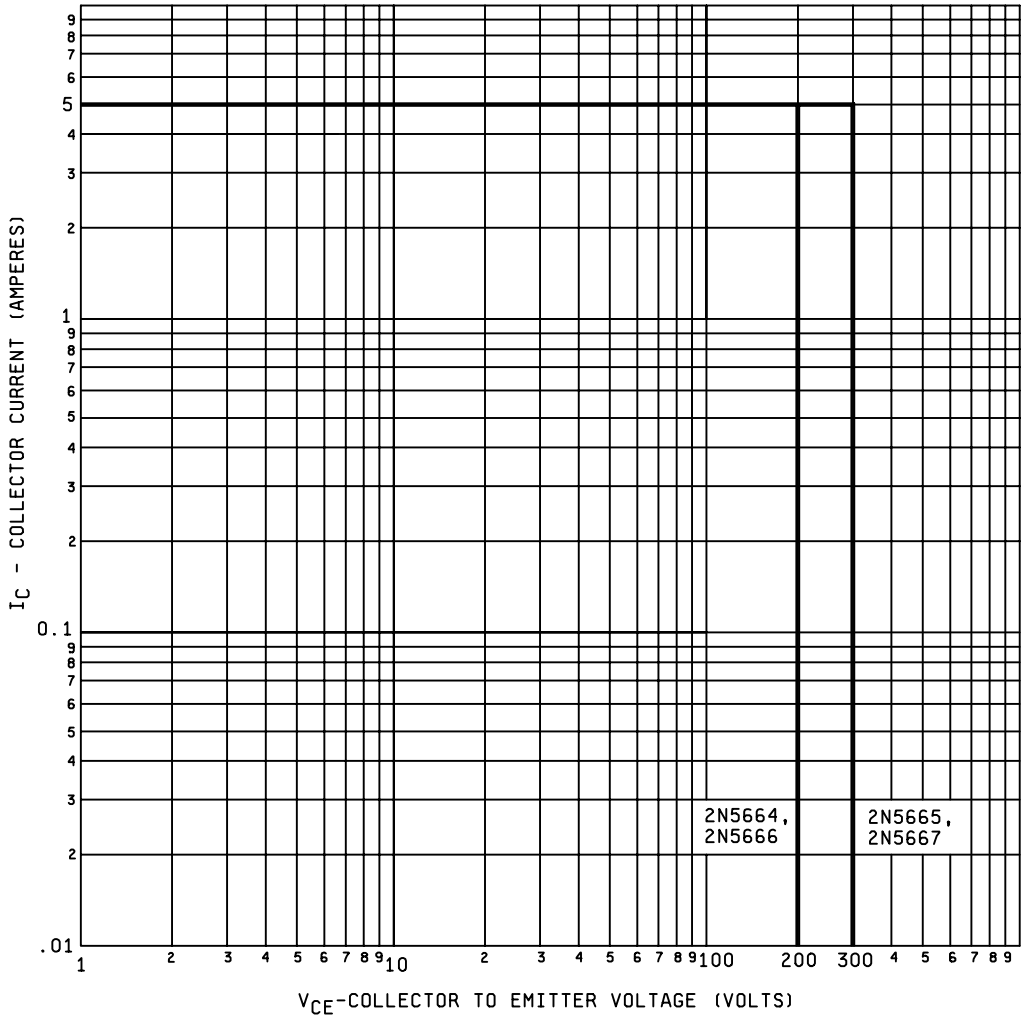
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NOTE: Electrical characteristics for "S" and "U3" suffix devices are identical to their corresponding devices without the suffix.

FIGURE 17. Maximum safe operating graph (continuous dc) for types 2N5666, 2N5666S, 2N5666U3, 2N5667, and 2N5667S.

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NOTE: Electrical characteristics for "S" and "U3" suffix devices are identical to their corresponding devices without the suffix.

FIGURE 18. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N5664) will be identified on the QML.

Die ordering information (1)	
PIN	Manufacturer
	43611
2N5664 2N5665 2N5666 2N5667	JANHCA2N5664 JANHCA2N5665 JANHCA2N5666 JANHCA2N5667

(1) For JANKC level, replace JANHC with JANKC.

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2897)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .

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